

APPAREL-LEDGER-V1: KHORA CHASSIS SPECIFICATION RECORD (REV 08)

MONOLITHIC V-PACM PRODUCTION & INTIMATE APPAREL SILHOUETTE ARCHITECTURE

1. Program Nomenclature & Identity Boundary

- Designation: Khora (Sub-variants: K-1 through K-6).
- Aesthetic Standard: Full conversion to everyday molded brassiere silhouette (contour T-shirt / plunge architecture). Industrial core elements are completely concealed within consumer intimate apparel form factors.
- Identity Shielding: Fully detached from

external entities, personal identifiers, regional registries, and proprietary naming prefixes. Zero cross-ledger attribution.

2. Tri-Layer Monolithic Fusion: Sequential Reaction Cast Molding (RLC / V-PACM)

The chassis utilizes Low-Pressure Vacuum-Assisted Sequential Cast Molding (V-PACM) with sacrificial water-soluble mandrels, integrating an apparel-grade feather-edge perimeter and direct textile co-molding.

[SEQUENTIAL GEL-CAST CHEMICAL TIMELINE]

STAGE 1: Cast Frosted Outer Sheath (0.3 mm down to 0.15 mm feathered rim) ↳ B-Stage Gelation (70–75% cure) STAGE 2: Seat Soluble Sacrificial Mandrel ↳ Vacuum Cast TPMS Core (Shore 30A–58A) STAGE 3: Seat Perimeter Textile Substrates + Cast Grooved Skin Interface (0.6 mm) ↳ Full Thermal Vulcanization (100% covalent cross-linking + mechanical textile lock)

STAGE 4: Closed-Loop Hot Water Wash ↴

Mandrel Dissolution & Core Channel Evacuation

- **Covalent Wet-on-Wet Vulcanization & Textile Co-Curing:**
 - Platinum-catalyzed addition-cure polydimethylsiloxane (LSR) layers are poured sequentially at the B-stage / "green state" (70–80% cure).
 - Unreacted vinyl ($-\text{CH}=\text{CH}_2$) and hydride ($\text{Si}-\text{H}$) groups cross layer boundaries before final thermal curing, forming a single covalent molecular network.
 - Lateral wing interfaces feature in-mold encapsulation of high-gauge micro-mesh textiles, achieving fiber lock directly during vulcanization without adhesive glue lines.
- **Sacrificial Core Mandrel Architecture:**
 - Open channels of the TPMS Gyroid are

formed using binder-jetted water-soluble mandrels (polyvinyl alcohol / synthetic ceramic salt matrix).

- Hot-water wash dissolves the internal mandrel through concealed lower drainage ports, leaving clean, smooth-walled open-cell passages.
- **Continuous In-Mold Durometer Blending:**
 - Zonal pouring into the casting tool transitions from Shore 50A–58A at the load-bearing lower shelf down to Shore 30A–35A at the upper apex without parting lines or physical seams.

3. Layer Cross-Sectional Geometry & Apparel Camouflage

[OUTSIDE: CLOTHING INTERFACE]

----- Layer 3: 0.3 mm Frosted-Matte Sheath (SPI A-3 / VDI 24) -> 0.15 mm Feathered Perimeter

----- Layer 2: Spatially Graded TPMS Gyroid Core
(Concealed Bellows Chamber)

----- Layer 1: Sinusoidal Drainage Ribs ($R = 0.8$
mm) + 35 mm Smooth Recessed Relief Zone

----- [INSIDE: EPIDERMAL CONTACT
INTERFACE]

- Layer 1 (Epidermal Contact Boundary):
 - Groove Geometry: Sinusoidal micro-ridges ($R = 0.8 \text{ mm}$) rounded crests, [1.2](#) mm width, 0.6 mm depth) directing fluid downward via capillary action and gravity.
 - Relief Zone: 35 mm smooth circular relief zone directly over the areola. Zero seams, zero lattice ridges, zero nipple abrasion.

- **Surface Treatment:** Micro-matte finish with permanently bonded oleophobic plasma monolayer ($\mu < 0.2$) to eliminate skin stiction and shear stress.
- **Layer 2 (Structural Metamaterial Core):**
 - **Geometry:** Continuous Triply Periodic Minimal Surface (Gyroid) open-channel sheet network with curved continuous boundaries resembling traditional molded contour cups.
 - **Fluid Dynamics:** 0.0% water/sebum absorption. Cyclic chest movement drives bellows-effect bidirectional air pumping ([1.2](#) to [7.6](#) L/min evacuation) venting out through concealed lateral edge vents.
 - **Sanitization:** Machine washable up to 60°C (140°F); zero detergent retention; spins dry instantly.
- **Layer 3 (Aesthetic T-Shirt Contour Shield):**

- Continuous, non-porous 0.3\text{ mm} elastomeric outer membrane tapering to a razor [0.15\text{ mm}](#) flange along the neckline and underarms to eliminate visible cup edges under clingy garments.
- Completely masks internal lattice structure with zero print-through beneath tight-knit modal, silk, or jersey outerwear.
- Matte microfiber-mimetic surface finish (VDI 24–27 / SPI A-3) matching standard luxury lingerie hand-feel.

4. Master 6-Core Tooling Matrix (K-1 to K-6)

Unit	Target	Band / Cup	IM F Radius	Zonal Duromete	Base Depth /	Apex Depth /	Biomechanical
	Volume						

	e (\text{cm}^3)	Eq uiv ale nt	(R_{\text{MF}})	r	Voi d %	Voi d %	& Ae sth etic Silh oue tte Fun ctio n
K-1	120 – 320	26A – 34B	95 – 120 mm	Shore 30A	1.8 mm / 60 %	0.6 mm / 90 %	Petite profile. Low-pro

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K-2	28 0 – 65 0	28 C – 36 D	11 0 – 13 5 m m	Sh ore 35 A	2.2 m m / 55 %	0.8 m m / 88 %	Sta nda rd roo t, nat ural for war d pro ject ion. Se aml

							ess eve ryd ay con tou r pro file.
K-3	55 0 – 95 0	30 DD – 38 E/ DD D	12 5 – 15 0 m m	Sh ore 42 A	2.6 m m / 50 %	0.9 m m / 85 %	Mi d- ma ss ele vati on. Bal con

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K-4	85 0 – 1,4 50	30 F – 40 G	14 0 – 16 5 m m	Sh ore 48 A	3.0 m m / 45 %	1.0 m m / 80 %	Hig h ma ss pro ject ion. Ca ntil eve red bas e she lf

							sho rte ns mo me nt ar m; sm oot h rou nde d pro ject ion.
K-5	1,3 50	32 GG	15 5	Sh ore	3.4 m	1.1 m	He avy

		– 2,0 00	– 44 J	– 18 0 m m	52 A	m / 40 %	m / 75 %	bus t (1. 5 – 2.5 kg) . Re pla ces ste el un der wir e cha
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K-6	1,9	34	17	Sh	3.8	1.2	Ma

	00 – 2,6 00 +	J – 48 M	0 – 20 0 m m	ore 58 A	m m / 35 %	m m / 70 %	xim um dyn ami c sup por t. Wi de win g dist rib uti on; eli mi nat
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5. Biomechanical Load Mitigation & Ergonomics

\text{Sagittal Bending Moment: } M = F_{\text{gravity}} \times d_{\text{anterior}}

- **85/15 Ribcage Load Distribution:**
 - Over 85% of total static and dynamic mass is grounded circumferentially into the rigid thoracic cage via the fabric-encapsulated Micro-Rail Foundation Band.
 - Shoulder straps transfer < 15% of vertical load, functioning strictly as positional stabilizers rather than primary suspension lines. Eliminates trapezius indentation and cervical spine (C7–T1) compression.
- **Moment Arm Shortening ($d_{\text{anterior}} \rightarrow d_{\text{min}}$):**
 - Cantilevered TPMS base shelves in Cores K-4, K-5, and K-6 lift and center breast tissue upward and inward toward the

sternal center of gravity, shortening the anterior moment arm distance (d) and reducing continuous forward rotational torque (M) by 35–40% in large cup profiles.

- **Continuous Hoop Stress Dispersion:**
 - Wide-profile silicone core encased in stretch micro-mesh eliminates localized band cutting and rolling, distributing compressive force uniformly across the ribs without soft-tissue bruising.
- **Costal Respiratory Freedom:**
 - Segmented flexure notches along the lower base shelf dynamically flex ($R_{\text{IMF}} = 95\text{ mm} - 200\text{ mm}$), accommodating full costal and diaphragmatic excursion during respiration without underwire stabbing.

6. Intimate Apparel Hardware & Concealed

Closure Architecture

- **Concealed Jewelry-Grade Front Clasp:**
 - Recessed N52 neodymium magnetic interlocking array encased in a satin-anodized or high-polish PVD low-profile metal housing at the sternal gore, visually identical to a high-end front jewelry closure.
 - Internal sliding dovetail carries horizontal and vertical shear forces ($> 180\text{ N}$), preventing pop-offs during movement while releasing with a single upward thumb flick.
- **Textile-Integrated Wings & Straps:**
 - Lateral wings feature ultra-flat bonded edges constructed from four-way stretch polyamide/elastane micro-mesh encapsulating the structural micro-rail.
 - Straps use low-profile slide hardware with

soft-brushed microfiber linings and internal non-stretch structural ribbons, eliminating the appearance of medical or athletic harness rigging.

- **Universal Donning:**
 - Front magnetic breakaway and flexible micro-rail band allow standard vest-style dressing or stepping into the chassis without requiring overhead arm extension.

7. Financial & Unit Economics Summary

- **Tooling CapEx:**
 - 6 Master Molds + Textile Overmold Dies + Extrusion Tooling + Clasp Enclosure:
\$158,000 – \$194,000 total upfront.
- **Unit Production Cost (COGS per Complete System):**
 - Pilot Production (1,000 units): \$28.50 / unit

- Scaled Production (50,000+ units): \$[13.40](#) / unit
- Retail Positioning & Margins (DTC):
 - Complete Chassis (2 Concealed Contour Cups + Micro-Mesh Rail Band + Jeweled Clasp): \$95.00 – \$115.00 (~86–88% gross margin at scale).
 - Modular Single Cup Replacement:\$38.00 (~84% gross margin).
 - Micro-Mesh Rail Foundation Band:\$32.00 (~87% gross margin).

8. Immediate Execution Roadmap

1. Tooling & Flange Revision: Update 3-piece CNC aluminum mold files for pilot unit K-3 with perimeter edge feathering down to [0.15](#) mm and SPI A-3 cavity polishing.
2. Textile Co-Molding Dwell Verification: Calibrate in-mold bonding parameters for

polyamide micro-mesh wings during Stage 3 vulcanization (85°C to 120°C ramp, 7-minute cycle) to confirm delamination resistance ($> 45 \text{ N/cm}$ peel strength).

3. **Apparel Clasp Enclosure Sampling:** Machine first-article satin PVD aluminum housings for the N52 dovetail core and run automated 2,000-cycle click-retention testing under dynamic loads.